
ENGINEERING TRIPOS PART IIA

ELECTRICAL AND INFORMATION ENGINEERING TEACHING LABORATORY

MODULE EXPERIMENT 3F1

FLIGHT CONTROL

Name: Liu Zihe
CRSID: zl559
College: Homerton
Date: 29 November 2025

Objectives:

- Simulation of various aircraft models on the computer.
- Study real-time (manual) control and the limitations imposed by time delays.
- Design of a simple autopilot.
- Illustrate frequency response concepts in analogue and digital control systems, conditions for oscillation in feedback systems and stability.
- Gain familiarity with Python and Jupyter Notebooks.

3F1 Lab Worksheet

2.1 Simplified Aircraft Model

$$\text{Transfer function} = \frac{10}{s^2 + 10s}$$

num = [10] den = [1, 10, 0]

2.2 Modelling Manual Control

$$\text{Controller transfer function} = K(s) = ke^{-sD}$$

$$k = 0.8 \quad D = 1 \text{ s}$$

$$\text{Phase margin} = 45 \text{ degrees}$$

$$\text{Amount of extra time delay which can be tolerated} = \frac{45 \times \pi / 180}{0.8} \approx 0.98 \text{ s (approx 1.0 s)}$$

2.3 Pilot Induced Oscillation

$$\text{Period of oscillation (observed)} = 5 \text{ s}$$

$$\text{Period of oscillation (theoretical)} = \frac{2\pi}{\omega_{pc}} = \frac{2\pi}{0.8} \approx 7.85 \text{ s}$$

2.4 Sinusoidal disturbances

$$\text{Maximum stabilising gain} = 4.5 \text{ (Gain Margin)}$$

$$\text{Gain at 0.66 Hz} = 5.5 \text{ dB}$$

$$\text{Phase at 0.66 Hz} = -90 \text{ degrees}$$

2.5 Unstable Aircraft

$$\text{Fastest pole at } T = 0.6$$

3.2 Autopilot with Proportional Control

$$\text{Proportional gain } K_c = 12$$

$$\text{Period of oscillation } T_c = 2 \text{ s}$$

3.3 Autopilot with PID Control

$$\text{Transfer function of PID controller} = K_p \left(1 + \frac{1}{T_i s} + T_d s \right)$$

$$\text{PID constants: } K_p = 7.2 \quad T_i = 1.0 \quad T_d = 0.25$$

$$\text{Adjusted value of } T_d = 0.35$$

3.4 Integrator Wind-up

$$\text{Integrator bound } Q = \frac{d \cdot T_i}{K_p} = \frac{2 \times 1.0}{7.2} \approx 0.278$$

3F1 Flight Control Lab Report

Question 1. (§2.2 Modelling Manual Control) Nyquist diagram (from Bode diagram) for controller in series with plant.

System Model: The simplified aircraft has transfer function $G(s) = \frac{N}{s^2 + Ms} = \frac{10}{s(s+10)}$, representing an integrator ($1/s$) in series with a first-order lag. The human controller is modelled as $K(s) = ke^{-sD}$ with proportional gain $k = 0.8$ and reaction delay $D = 1$ s.

Open-Loop Transfer Function:

$$L(s) = K(s)G(s) = \frac{8e^{-s}}{s(s+10)}$$

Frequency Response: Substituting $s = j\omega$:

$$L(j\omega) = \frac{8e^{-j\omega}}{j\omega(j\omega+10)} = \frac{8}{j\omega \cdot \sqrt{\omega^2+100}} \cdot e^{j \arctan(\omega/10)} \cdot e^{-j\omega}$$

The magnitude and phase are:

$$|L(j\omega)| = \frac{8}{\omega\sqrt{\omega^2+100}}, \quad \angle L(j\omega) = -\frac{\pi}{2} - \arctan\left(\frac{\omega}{10}\right) - \omega$$

Key Points from Bode Diagram (Figure 2):

- **Low frequency** ($\omega \rightarrow 0$): $|L| \rightarrow \infty$, $\angle L \rightarrow -90^\circ$ (integrator dominates).
- **Gain crossover** ($\omega_{gc} \approx 0.8$ rad/s): $|L| = 1$ (0 dB). Phase at this point:

$$\angle L(j \cdot 0.8) = -90^\circ - \arctan(0.08) - 0.8 \times \frac{180^\circ}{\pi} \approx -90^\circ - 4.6^\circ - 45.8^\circ \approx -140^\circ$$

Phase margin = $180^\circ - 140^\circ = 40^\circ$ (approximately 45° from graph).

- **High frequency** ($\omega \rightarrow \infty$): $|L| \rightarrow 0$, $\angle L \rightarrow -\infty$ (time delay adds unbounded phase lag $-\omega$ rad).

Constructing Nyquist from Bode: The Nyquist plot traces $L(e^{j\theta})$ as ω varies from 0 to ∞ . Starting from the negative imaginary axis (phase -90° , large magnitude), the locus spirals clockwise towards the origin as magnitude decreases and phase becomes increasingly negative. The critical point -1 is not encircled, confirming closed-loop stability.

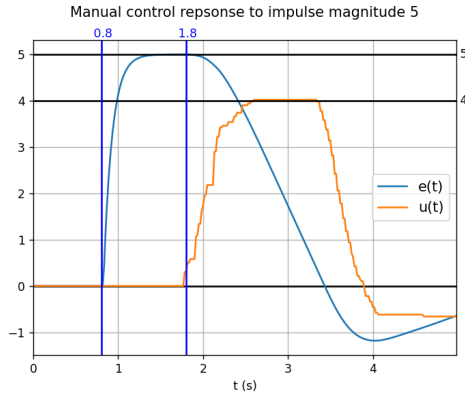


Figure 1: Manual control response to impulse disturbance. From the plot: $k \approx u_{max}/e_{max} = 4/5 = 0.8$, $D \approx 1$ s (delay before response).

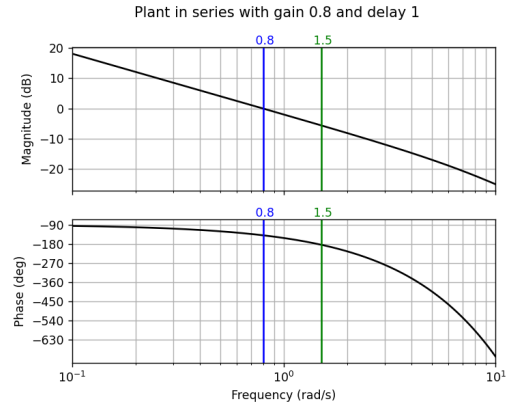
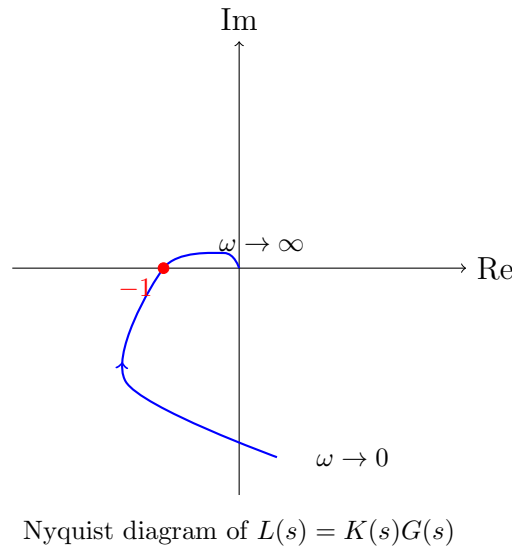


Figure 2: Bode diagram: open-loop $L(s)$ with $k = 0.8$, $D = 1$ s. Phase margin $\approx 45^\circ$ at $\omega_{gc} = 0.8$ rad/s.



2. (§2.2 Modelling Manual Control) Are you using any integral action?

Yes. In Figure 3, the manual control input $u(t)$ does not return to zero but maintains a steady value to counteract the step disturbance, while $e(t) \rightarrow 0$. This implies the human controller provides integral action (“trimming”) to eliminate steady-state error. The simple model $K(s) = ke^{-sD}$ is therefore an approximation that neglects the pilot’s low-frequency adaptation.

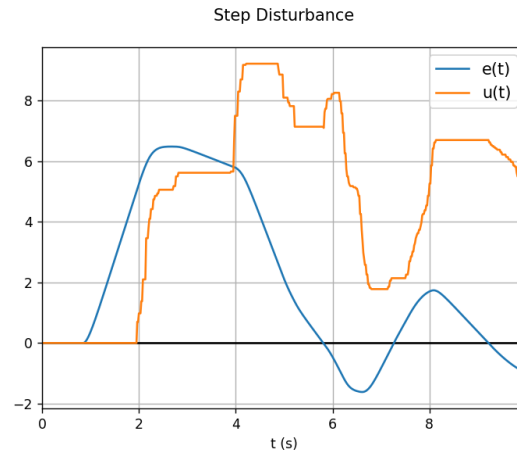


Figure 3: Step disturbance response showing integral action: $u(t)$ maintains steady offset to drive $e(t) \rightarrow 0$.

3. (§2.3 Pilot Induced Oscillation) Explain the oscillation of the feedback loop.

Mechanism: PIO occurs when the pilot's time delay introduces sufficient phase lag that the total loop phase reaches -180° at a frequency where the loop gain exceeds unity. The PIO aircraft has $G_1(s) = c/(Ts + 1)^3$ which contributes up to -270° phase lag, plus the pilot delay e^{-sD} adds $-\omega D$ radians.

Stability condition: At the phase crossover frequency ω_{pc} where $\angle L(j\omega_{pc}) = -180^\circ$, if $|L(j\omega_{pc})| > 1$, the system is unstable and oscillates at frequency ω_{pc} .

Period comparison: From Bode (Figure 6), $\omega_{pc} \approx 0.8$ rad/s, giving theoretical period $T = 2\pi/\omega_{pc} \approx 7.85$ s. The observed period (≈ 5 s) is shorter, suggesting the pilot adapts their gain/delay under oscillatory conditions.

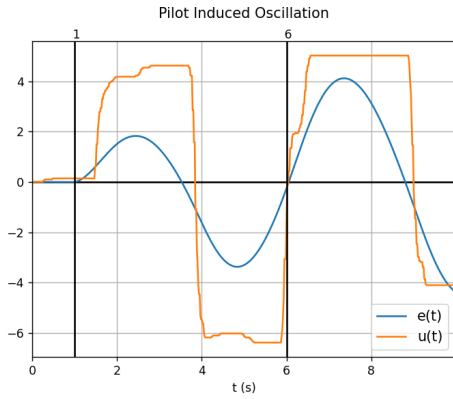


Figure 4: PIO with manual control. Period ≈ 5 s.

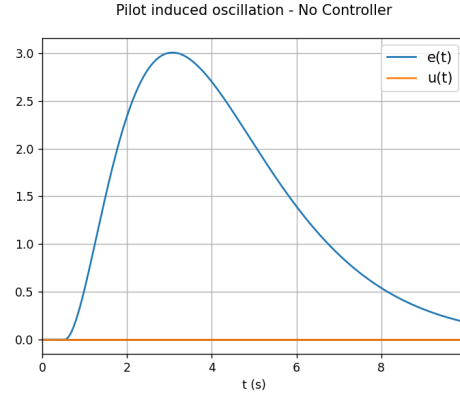


Figure 5: No control: aircraft is naturally stable.

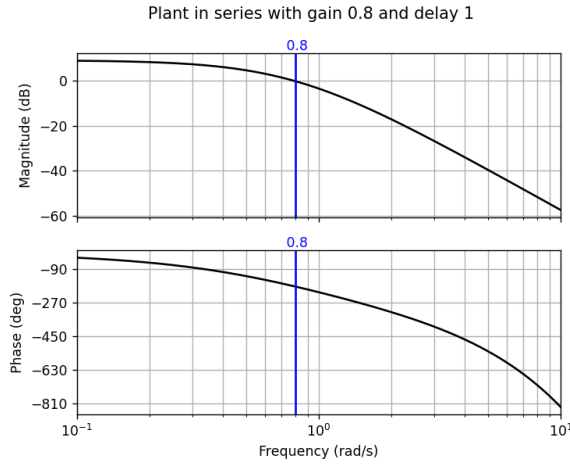


Figure 6: PIO Bode diagram. Phase crossover at $\omega_{pc} \approx 0.8$ rad/s.

4. (§2.3 Pilot Induced Oscillation) Can you give a rough guideline to the control designer to make PIO less likely?

To prevent PIO, ensure adequate phase margin at frequencies where the pilot is active (< 1 Hz):

- Reduce total system delay (sensors, actuators, displays)
- Avoid plants with high phase lag at low frequencies

- Design for phase margin $> 45^\circ$ at the expected pilot crossover frequency
- Consider rate limiters to prevent high-frequency pilot inputs

5. (§2.4 Sinusoidal Disturbances) Was your manual input able to reduce the error?

No. The closed-loop response to disturbance d is $y = \frac{G}{1 + KG}d$. For disturbance rejection, we need $|1 + KG| > 1$.

At 0.66 Hz ($\omega = 4.15$ rad/s), pilot delay adds phase $-\omega D = -4.15 \times 1 \approx -238^\circ$. Combined with plant phase ($\approx -90^\circ$ from Bode), total phase approaches -330° , meaning KG points nearly opposite to -1 . This makes $|1 + KG| < 1$, so the disturbance is actually *amplified* rather than attenuated. Manual control worsens the oscillation.

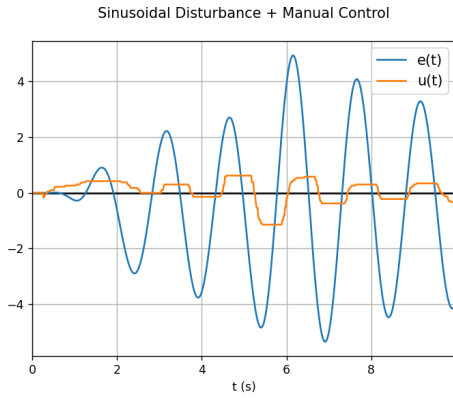


Figure 7: Sinusoidal + manual control.

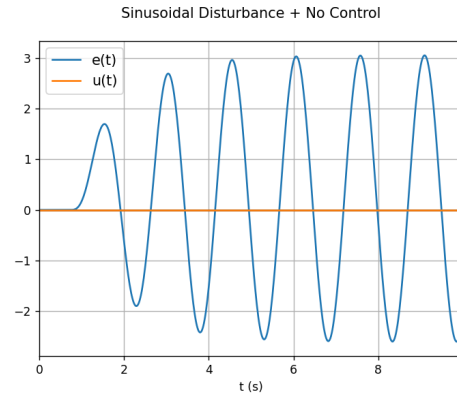


Figure 8: Sinusoidal + no control.

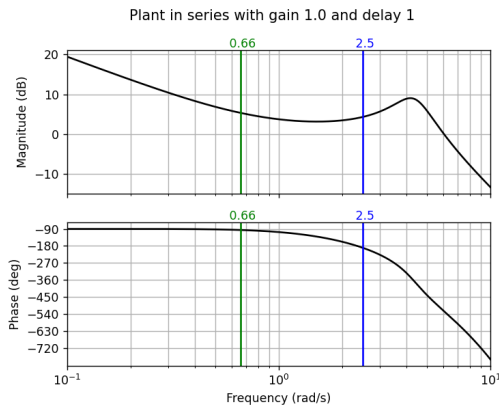


Figure 9: F4E Bode: gain margin 4.5; at 0.66 Hz: 5.5 dB, -90° .

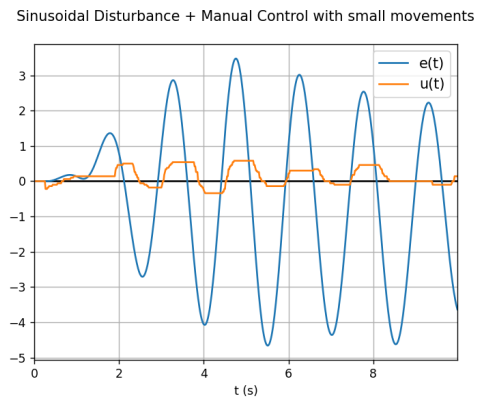


Figure 10: Small movements: still ineffective.

6. (§2.5 An Unstable Aircraft) Nyquist diagram for $G_2(s)$.

$G_2(s) = \frac{2}{sT - 1}$ has one unstable pole at $s = 1/T > 0$. The frequency response is:

$$G_2(j\omega) = \frac{2}{j\omega T - 1} = \frac{-2(1 + j\omega T)}{1 + \omega^2 T^2}$$

At $\omega = 0$: $G_2(0) = -2$. As $\omega \rightarrow \infty$: $G_2 \rightarrow 0$. The locus is a circle centered at $(-1, 0)$ with radius 1, traced clockwise from -2 to 0 .

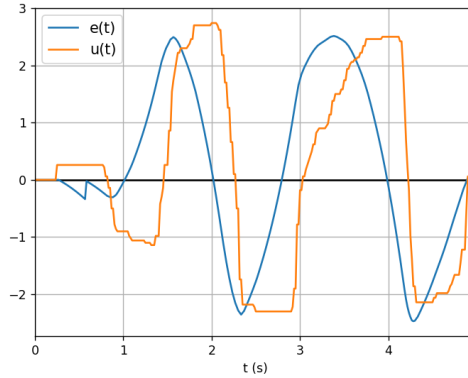
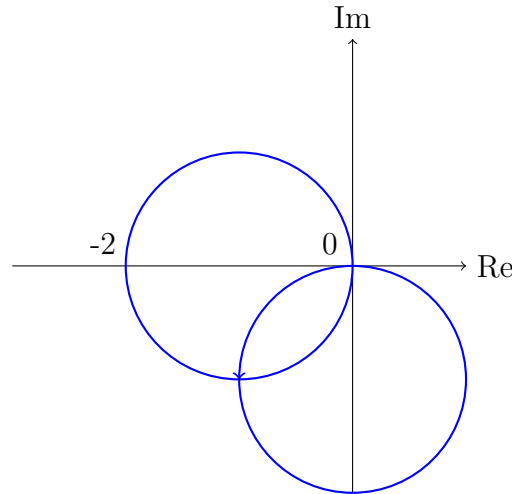


Figure 11: Unstable aircraft ($T = 0.6$) manually stabilized. Oscillatory near stability limit.



Nyquist diagram of $G_2(s)$

7. (§2.5 An Unstable Aircraft) Explain, using the Nyquist criterion, why the feedback system is stable with a proportional gain greater than 0.5.

Nyquist Criterion: For a system with P open-loop poles in the RHP, closed-loop stability requires $N = -P$ encirclements of $-1/K$, where $N > 0$ is counter-clockwise.

Here $P = 1$ (one unstable pole), so we need $N = -1$: one *clockwise* encirclement of $-1/K$.

The Nyquist locus is a circle from -2 to 0 on the real axis, traversed clockwise. For $-1/K$ to be encircled clockwise:

$$-2 < -\frac{1}{K} < 0 \implies 0 < \frac{1}{K} < 2 \implies K > 0.5$$

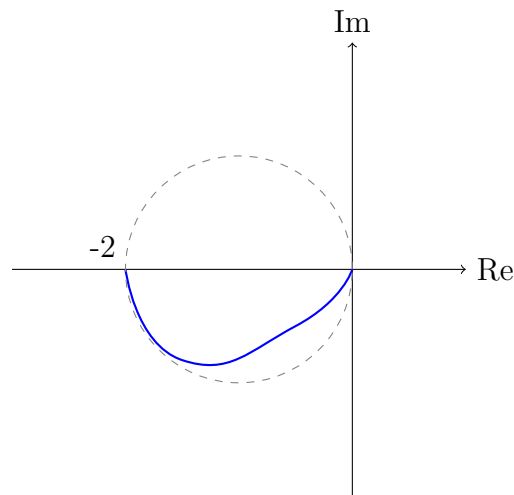
For $K > 0.5$, the point $-1/K$ lies inside the circle, giving the required clockwise encirclement for stability.

8. (§2.5 An Unstable Aircraft) Sketch of a Nyquist diagram for $G_2(s)$ with a small time delay D .

With delay, the transfer function becomes $G_2(s)e^{-sD}$. The delay contributes phase $-\omega D$ without changing magnitude:

$$|G_2(j\omega)e^{-j\omega D}| = |G_2(j\omega)|, \quad \angle(G_2e^{-j\omega D}) = \angle G_2 - \omega D$$

The additional phase lag causes the circular locus to spiral inward rather than close cleanly at the origin. For large D , the spiral may cross $-1/K$ multiple times, potentially destabilizing the system.



Nyquist diagram with delay (Solid)

9. (§3.4 Integrator Wind-up) Explain how you calculated the bound on Q .

PID Controller: $u(t) = K_p \left(e + \frac{1}{T_i} \int_0^t e \, d\tau + T_d \frac{de}{dt} \right)$

Steady-state analysis: After settling, $e_{ss} = 0$ and $\dot{e} = 0$. To cancel a step disturbance d , we need $u_{ss} = d$. Only the integral term contributes:

$$u_{ss} = \frac{K_p}{T_i} I_{ss} = d \implies I_{ss} = \frac{d \cdot T_i}{K_p}$$

Minimum bound: For $d = 2$, $T_i = 1.0$, $K_p = 7.2$:

$$Q = \frac{2 \times 1.0}{7.2} = \frac{5}{18} \approx 0.278$$

Setting Q smaller would prevent full disturbance rejection (steady-state error). Setting Q larger allows more wind-up overshoot.

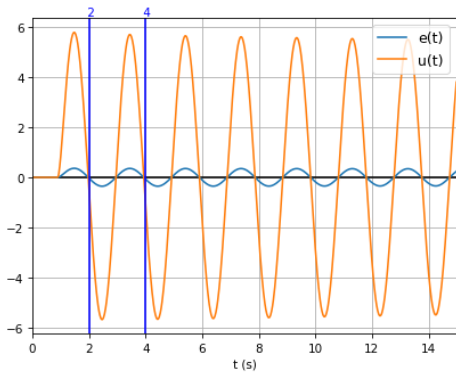


Figure 12: Proportional control: $K_c = 12$, $T_c = 2s$.

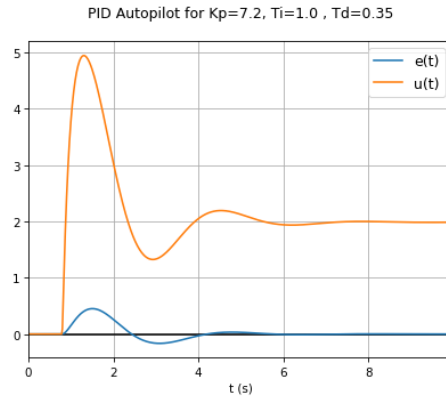


Figure 13: PID with $T_d = 0.35$: reduced oscillation.

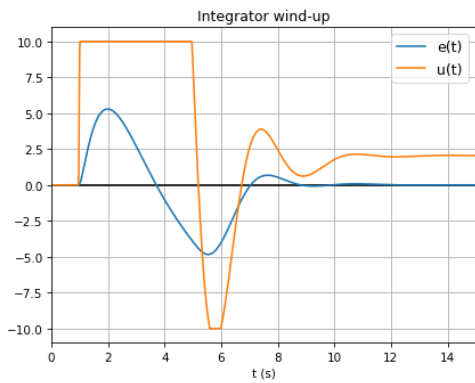


Figure 14: Integrator wind-up: large overshoot.

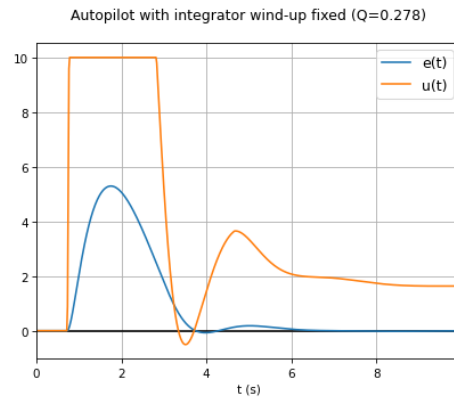


Figure 15: Wind-up fixed with $Q = 0.278$.